Comparative Machine Learning for Spam Detection

Evaluating four classifiers using TF-IDF features

01. The Problem

User Exposure

Unwanted advertising and malicious scams.

Binary Classification

Distinguish legitimate ham from unwanted spam.

Algorithm Comparison

Evaluate four standard models using identical TF-IDF features.

02. Methodology Pipeline



Dataset

5,171 raw messages.



Clean Text

Remove punctuation, lemmatization, and stopwords.



TF-IDF

Max 5,000 text features.



Split

80/20 stratified division.

02. Evaluated Classifiers

Naive Bayes

Probabilistic baseline for text.

Logistic Regression

Linear class probability estimation.

Random Forest

Ensemble of decision trees.

LinearSVC

High-dimensional margin separation.

03. Results: Test Set Accuracy



03. Results: Error Trade-offs

Logistic Regression

10

False Positives

4

False Negatives

LinearSVC

7

False Positives

7

False Negatives

Random Forest

16

False Positives

1

False Negatives

04. Key Takeaways

Linear Dominance

Logistic Regression and LinearSVC excel on this TF-IDF dataset.

Error Trade-offs

Application dictates acceptable false-positive versus false-negative rates.

Study Limitations

Undocumented dataset provenance and unused SMOTE balancing.